



Gym Space Management System (GSMS) T09

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DEMO Roadmap:

Introduction

RDD

SRS

SAD

Objectives

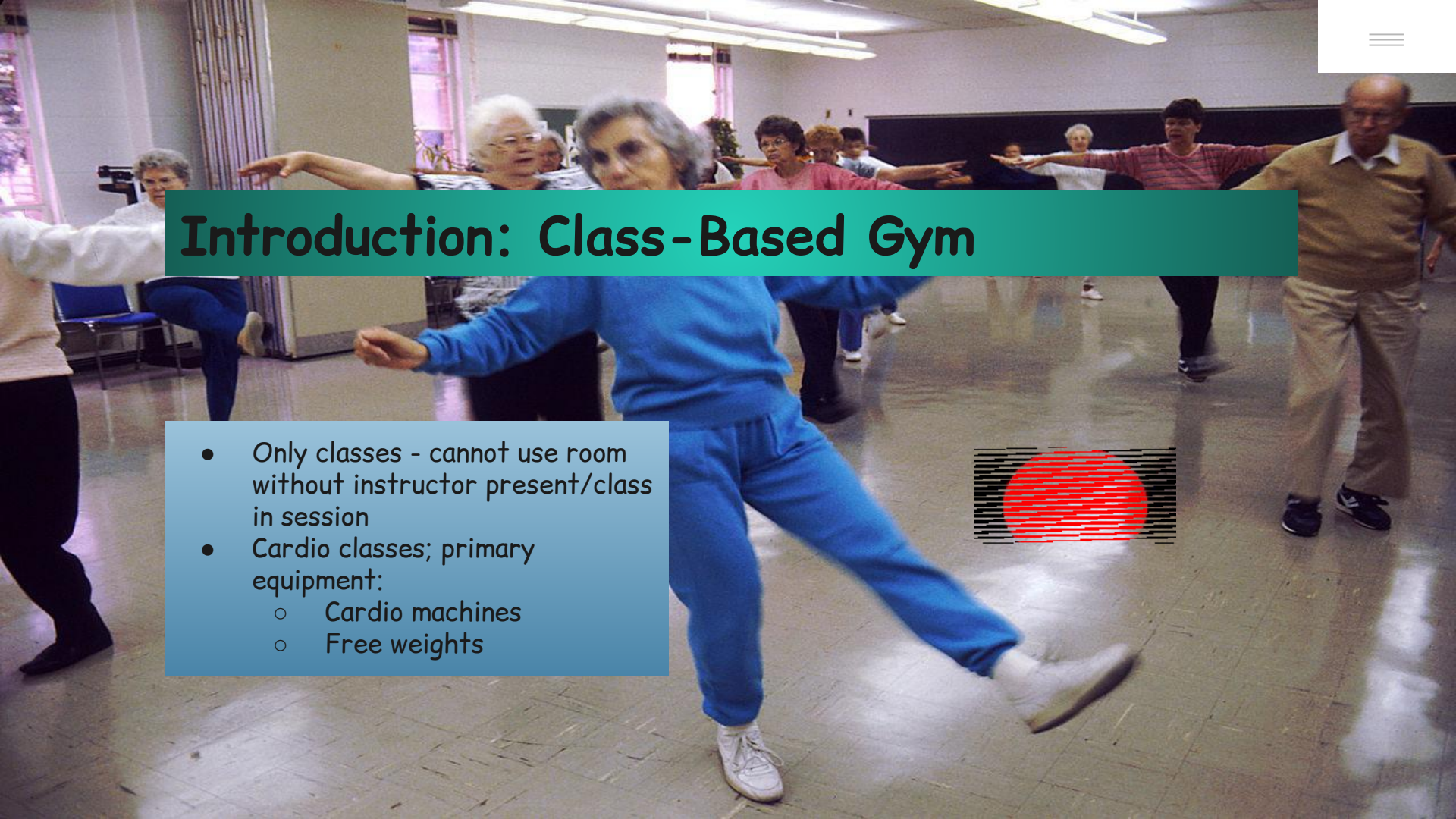
OMT

Diagram

Capabilities

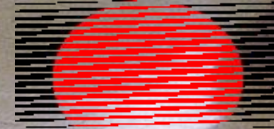
Use Cases





Introduction: Class-Based Gym

- Only classes - cannot use room without instructor present/class in session
- Cardio classes; primary equipment:
 - Cardio machines
 - Free weights



Objectives

1. **Improve Gym Safety:** Collect relevant health and fitness data and keeps members & instructors informed of unsafe activities.
2. **Enhance Community Fitness & Member Retention:** Observes & provides information to members to improve balanced exercise routines, as well as collecting & providing feedback on activities in the gym to staff & and instructors.
3. **Improve Facility Usage:** Provides analytics regarding resource usage and gym activities.



Capabilities

1

Real-Time Health & Safety Updates:

- Active monitoring and logging.
- Communication with degree of severity.
- Warns and Alerts involved parties during gym activities.

2

Personalized Fitness Feedback:

- Generating recommendations based on health and fitness logs.
- Access to relevant information by designated parties with sufficient permissions.

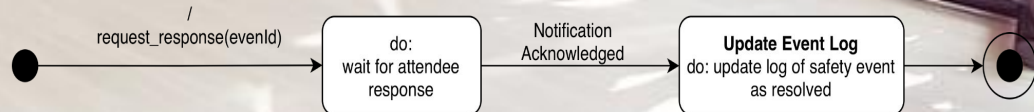
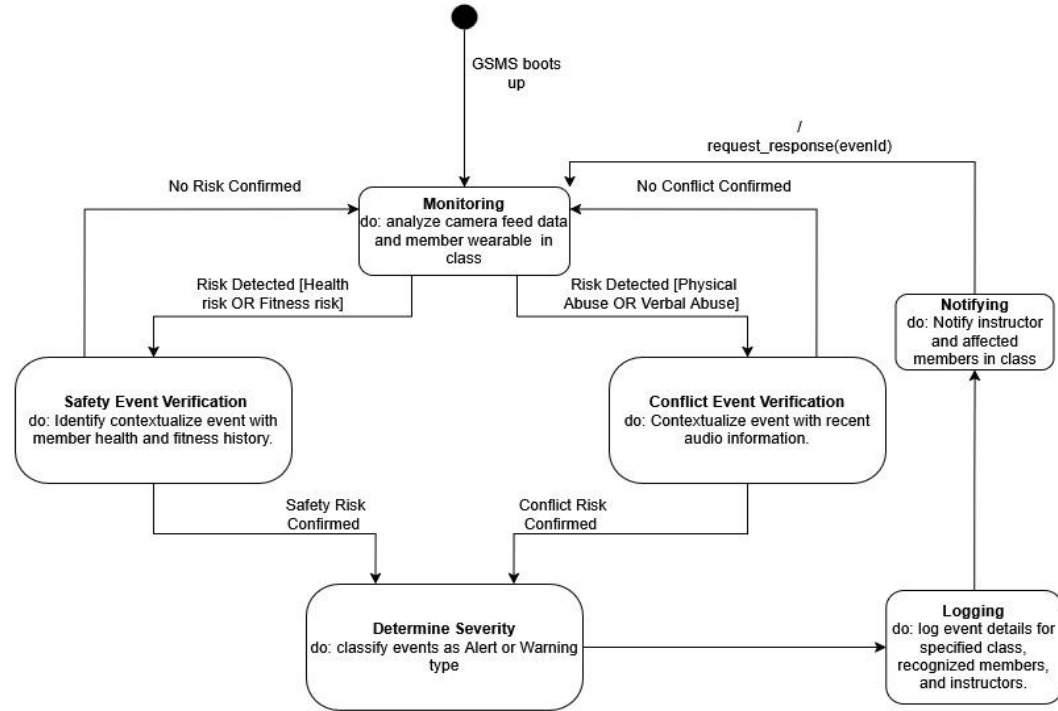
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Health, Fitness, & Gym History Storage:

- Safely stores member information.
- Tracks and logs activities in gym.
- Alerts instructors of unusual member activities.
- Generates a report.



OMT: Safety and Conflict Notification

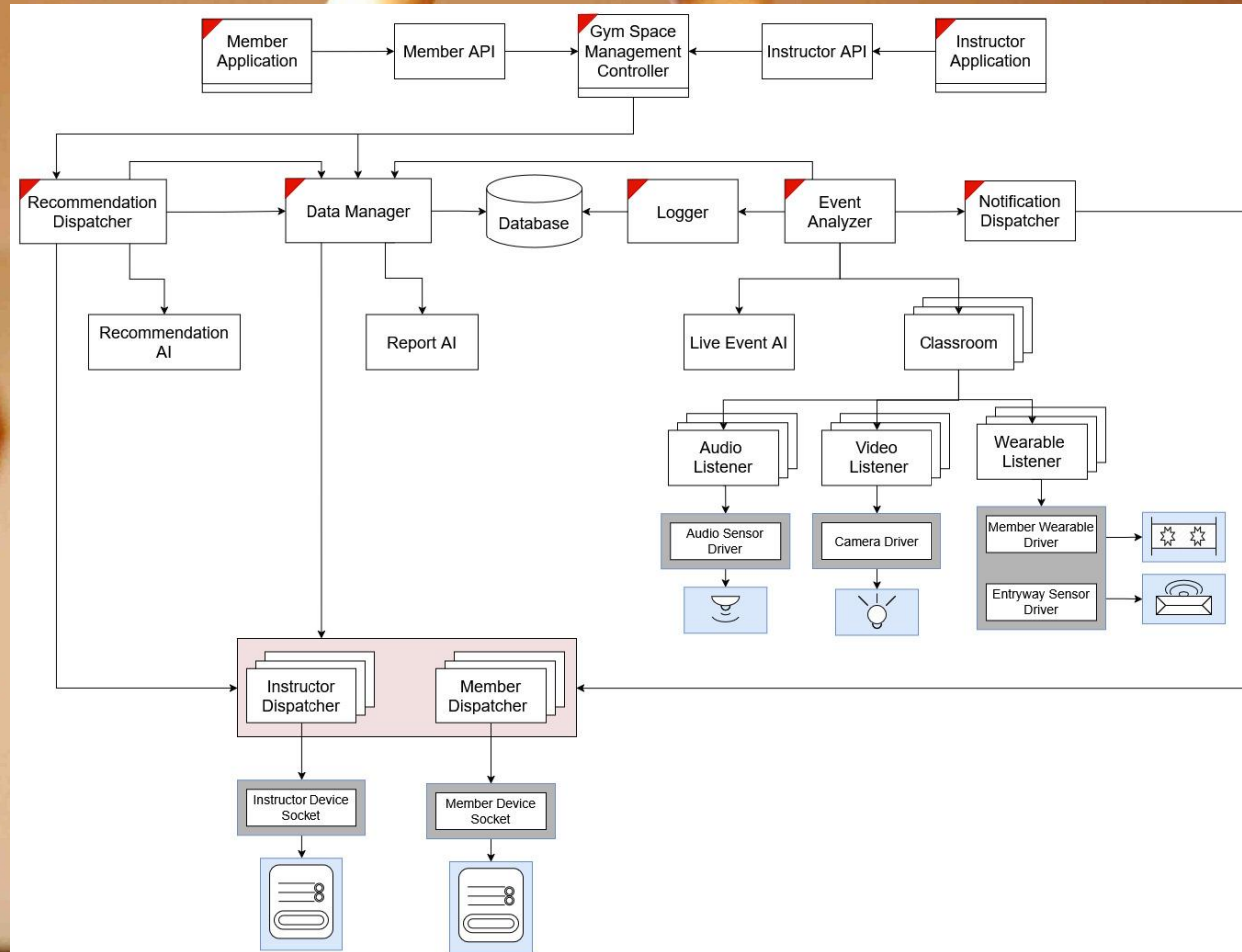


SAD

GSMS Architecture Overview

Takeaways:

- Responses to applications via wireless network
- Mixture of Event-Driven and Microservice-like architecture



Use Cases

Use cases presented in order of the demonstration scenarios.

- 01 | Generate a Recommendation for Auxiliary Workout
- 02 | Resolving Exhaustion in Cardio Class.
- 03 | Conflict Detection and De-escalation
- 04 | Detecting and Responding to a Member Health Emergency
- 05 | Generating a Monthly Usage PDF Report





Let's get to the DEMO!